

The ⟨kta⟩ Process Map

Proof of work for a generator-backed Sanskrit grammar archive

Notation

Form	Convention
<i>dhātu, gaṇa, niṣṭhā, kartari</i>	Sanskrit terms are italic IAST.
√ <i>gam</i>	The root sign marks a verbal root.
⟨ <i>kta</i> ⟩	Angle brackets mark an affix or grammatical form.
AS-8-2-42	Stable identifier for an Aṣṭādhyāyī rule.

Decision flow

1. Dhātu identity	Resolve the exact <i>Dhātupāṭha</i> entry, <i>gaṇa</i> , markers, meaning, preverb, and intended syntax.
2. Affix selection	Introduce ⟨ <i>kta</i> ⟩ and register it as <i>niṣṭhā</i> .
3. Voice and meaning	Test the ordinary resultative reading and any licensed <i>kartari</i> interpretation.
4. iṭ decision	Apply the <i>ārdhadhātuka iṭ</i> system; do not infer <i>seṭ</i> / <i>aniṭ</i> / <i>veṭ</i> from the <i>gaṇa</i> alone.
5. Root operations	Apply substitutions, augments, and ordered phonology.
6. Niṣṭhā rules	Check 8.2.42ff. for substitutions, lexical outcomes, options, and prohibitions.
7. Sandhi and audit	Record the final form, competing licensed outputs, sources, and unresolved questions.

Repository architecture

<i>Preamble 1</i>	Complete <i>Dhātupāṭha</i> registry with stable root anchors.
<i>Preamble 2</i>	Pāṇinian <i>sūtra</i> registry in numerical order.
Chapters 1-11	<i>Gaṇa</i> -specific derivation bars and local rule-use anchors.
Cross-cutting registries	<i>Sandhi</i> , composition categories, exceptions, and audit status.
Final proof	Compiled PDFs under <i>vyakaranamlikhitam</i> / <i><project></i> /.

Initial rule backbone

Each Sanskrit rule is presented in full IAST and separated from its English operational description.

Rule	Sanskrit sūtra - IAST	Operational use
1.1.26	<i>ktaktavatū niṣṭhā</i>	Defines ⟨kta⟩ and ⟨ktavatu⟩ as niṣṭhā.
3.2.102	<i>niṣṭhā</i>	Introduces the niṣṭhā affixes in the completed-action domain.
3.4.72	<i>gatyarthākarmakaśliṣaśīnsthāsavasajanaruhaḥ ryatibhyaś ca</i>	Licenses kartari interpretation for motion roots, intransitives, and listed roots.
7.2.35	<i>ārdhadhātukasyed valādeḥ</i>	Provides iṭ before a val-initial ārdhadhātuka affix, subject to later rules.
8.2.42	<i>radābhyām niṣṭhāto naḥ pūrvasya ca daḥ</i>	Begins central niṣṭhā substitutions after roots ending in r or d.
8.2.51	<i>śuṣaḥ kaḥ</i>	Prescribes the special k outcome for √śuṣ.
8.2.52	<i>paco vaḥ</i>	Prescribes the special v outcome for √pac.
8.2.56	<i>nudavidondatrāghrāhrībhyo 'nyatarasyām</i>	Allows optional treatment for the listed roots.
8.2.57	<i>na dhyākhyāp̐mūrchimadām</i>	Blocks the inherited operation for the listed roots.
8.4.55	<i>khari ca</i>	Registers a relevant final-devoicing environment.

Stable identifiers

DHATU-01-0001

AS-8-2-42

G01-AS-8-2-42-U01

G01-DHATU-01-0001-KTA

G01-SANDHI-AS-8-4-55-U01

Coverage status

This proves the implemented architecture; it does not claim that every Sanskrit root already has a fully audited ⟨kta⟩ derivation.

Sample derivational categories

Sanskrit is consistently marked as italic IAST; roots carry √ and affixes use angle brackets.

Category	Example	Audit note
Regular surface ⟨kta⟩	√bhū + ⟨kta⟩ → bhūta	High-frequency model with visible -ta.
Kartari motion-root reading	√gam + ⟨kta⟩ → gata	Motion roots can license a kartari interpretation under 3.4.72.
Lexical outcome	√pac + ⟨kta⟩ → pakva	The result depends on a specific niṣṭhā rule, not simple concatenation.
Optional/meaning-conditioned	√vid → vitta / vidita	Optionality and lexical meaning must be recorded separately.

Publication rule

Every exploratory grammar project ends with one or more human-inspectable PDFs in vyakaranamlikhitam/<project>/. Editable sources and generators remain elsewhere.